WORKED SOLUTION

STEP I 2007, Question 10

Mechanics • Displacement-Time Graphs

STEP I 2007, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q10

The key idea

On a displacement–time graph every rider and every army is a straight line (or two straight segments), and "passing each other" simply means two lines crossing. So we write down the equation of each line, measuring displacement from the Saxon army, and intersect them. The inequalities in the question are exactly the conditions that the crossings happen in the order we expect, and the final part is about what goes on when one of them fails.

Solution

Setting up the graph

We model the armies and horsemen as points moving along a straight line. Let s be the displacement from the (stationary) Saxon army, towards the Normans, and start the clock when the armies are d apart. Then:

- Saxon army: $s = 0$.
- Norman army: $s = d - ut$.
- Saxon horseman, outward: $s = xt$. He reaches the Norman army when $xt = d - ut$, so at $t_S = \dfrac{d}{x+u}$, at $s = \dfrac{xd}{x+u}$. Coming back at speed x,

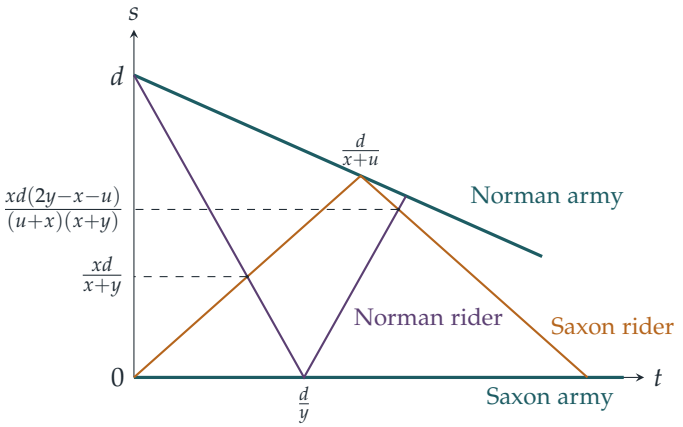
$$s = \frac{xd}{x+u} - x\left(t - \frac{d}{x+u}\right) = \frac{2xd}{x+u} - xt,$$

  and he is home at $t = \dfrac{2d}{x+u}$.
- Norman horseman, outward: $s = d - yt$. He reaches the Saxon army at $t_N = \dfrac{d}{y}$, and

coming back at speed y his line is

$$s = y\left(t - \frac{d}{y}\right) = yt - d.$$

Since $y > u$ he does catch his own army, when $yt - d = d - ut$, i.e. at $t = \dfrac{2d}{y+u}$.



(The picture is drawn with $u = 1$, $x = 2$, $y = 4$, $d = 1$.)

The first passing

Both riders are on their outward journeys, so we intersect $s = xt$ with $s = d - yt$: $(x+y)t = d$, giving

$$t = \frac{d}{x+y}, \qquad s = \frac{xd}{x+y}.$$

This is genuinely before either has turned round, because $\frac{d}{x+y}$ is less than both $\frac{d}{y}$ and $\frac{d}{x+u}$ (using $y > u$ for the second).

The second passing

Now both riders are on their way home: we intersect $s = \dfrac{2xd}{x+u} - xt$ with $s = yt - d$. Adding $xt + d$ to both sides,

$$(x+y)t = d + \frac{2xd}{x+u} = \frac{d(3x+u)}{x+u}, \qquad \text{so} \qquad t_2 = \frac{d(3x+u)}{(x+u)(x+y)}.$$

Then

$$s = \frac{2xd}{x+u} - \frac{xd(3x+u)}{(x+u)(x+y)} = \frac{xd\big(2(x+y) - (3x+u)\big)}{(x+u)(x+y)} = \frac{xd(2y-x-u)}{(u+x)(x+y)},$$

which is the second distance required.

Ask yourself How do we know both riders really are on their return legs at time t_2?

This is where the inequalities come in. The Norman rider has turned if $t_2 \geq \frac{d}{y}$:

$$y(3x+u) \geq (x+u)(x+y) \iff 3xy \geq x^2+xy+ux \iff u \leq 2y-x.$$

The Saxon rider has turned if $t_2 \geq \frac{d}{x+u}$:

$$3x+u \geq x+y \iff u \geq y-2x.$$

So the condition $y-2x < u < 2y-x$ is exactly what makes t_2 come after both turning points. The same condition $u < 2y-x$ also makes $s > 0$, so the Saxon rider is not yet home. (There is one more, unstated, requirement: the Norman rider must not have reached his own army first. Working through $t_2 \leq \frac{2d}{y+u}$ reduces to $(x-u)(2x-y+u) \geq 0$, and since $2x-y+u > 0$ this means $x \geq u$, i.e. the Saxon rider is at least as fast as the Norman army. That is surely intended: otherwise, the moment he turned round, the army would overtake him.)

Case (i): $u > 2y-x$

Now $t_2 < \frac{d}{y}$: the lines for the Saxon rider's return and the Norman rider's *outward* journey cross first. Since $y > u$, this case needs $x > y$: the Saxon rider is the faster one. He reaches the Norman army, turns, and catches up with the Norman rider from behind while both are heading towards the Saxon army. (Intersecting $s = \frac{2xd}{x+u} - xt$ with $s = d - yt$ puts this at $\frac{xd(x+u-2y)}{(x+u)(x-y)}$ from the Saxon army.) He is home at $\frac{2d}{x+u}$, before the Norman rider arrives at $\frac{d}{y}$, so the two do not meet head-on a second time: the Norman rider simply arrives at the Saxon army, where his opponent is already back.

Case (ii): $u < y-2x$

Now $t_2 < \frac{d}{x+u}$: the Norman rider is much faster. He reaches the Saxon army, turns, and overtakes the Saxon rider, who is still riding out towards the Normans. Intersecting $s = yt - d$ with $s = xt$, this happens at $t = \frac{d}{y-x}$, at $\frac{xd}{y-x}$ from the Saxon army. The Norman rider then gets home at $\frac{2d}{y+u}$, before the Saxon rider reaches the Norman army at $\frac{d}{x+u}$ (because $2(x+u) < y+u$), and there is no further passing on the open field.

Checks

- With $d=1$, $u=1$, $x=2$, $y=4$ (as in the picture), the formulae give passings at $\frac{1}{3}$ and $\frac{5}{9}$. Stepping the motion numerically in time steps of 10^{-5} gives crossings at 0.3333 and 0.5556.
- If $y = x+u$ the two distances are equal, both being $\frac{x}{x+y}d$. That is a nice special value to test.

- As $u \to 2y - x$ the second distance tends to 0: the second passing happens right at the Saxon army, which is the boundary with case (i).

Common slips

- Writing the return leg of a rider as $s = -xt$ or $s = yt$, forgetting to shift the time origin to the moment of turning.
- Using the Norman army's position at $t = 0$, rather than at the moment the Saxon rider arrives, for his turning point.
- In the final part, saying only that "they do not meet". In both cases they do pass a second time, but going the same way: one overtakes the other.

Beyond STEP

Graphs like this are *world lines*, the basic tool of special relativity. There, light signals sent back and forth between moving observers are analysed in exactly this way (the radar method), and the slopes of the lines encode the relative velocities. The algebra here, with u added to one speed and subtracted from another, is Galilean relative velocity. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers both the Galilean and the relativistic pictures.

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